Computer Networks

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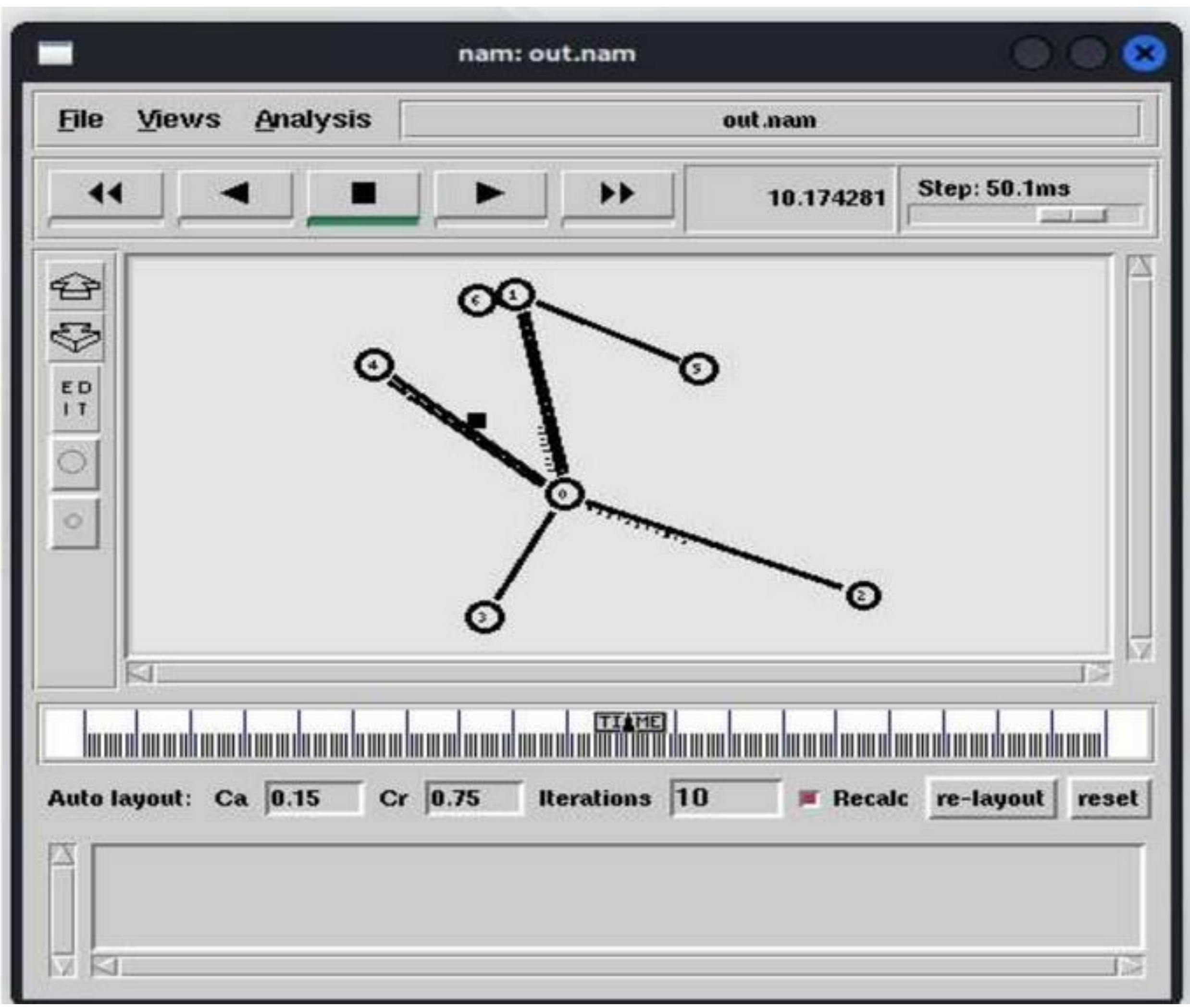
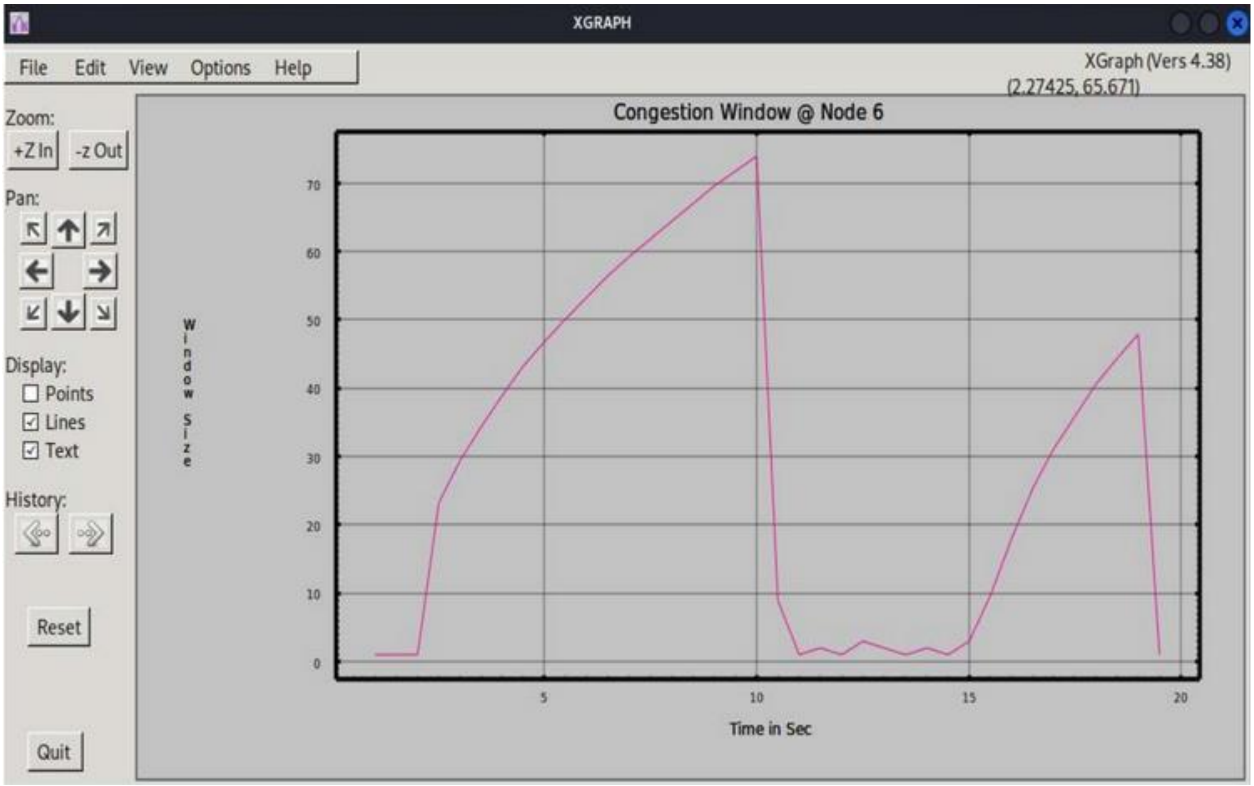
Section: H

Topic:
NS2 Congestion Window

Packet Drop



XGraph



Explanation:

X-axis (Time in Sec): Represents the passage of time in seconds.

Y-axis (Window Size): Represents the size of the congestion window (CWND), indicating how many packets the TCP sender can transmit before waiting for an acknowledgment.

Slow Start (0 - 10 seconds):

Initially, the congestion window size increases exponentially. This phase represents TCP's *slow start*, where the window size grows quickly as the sender aggressively tries to increase throughput.

Congestion Event (~10 seconds):

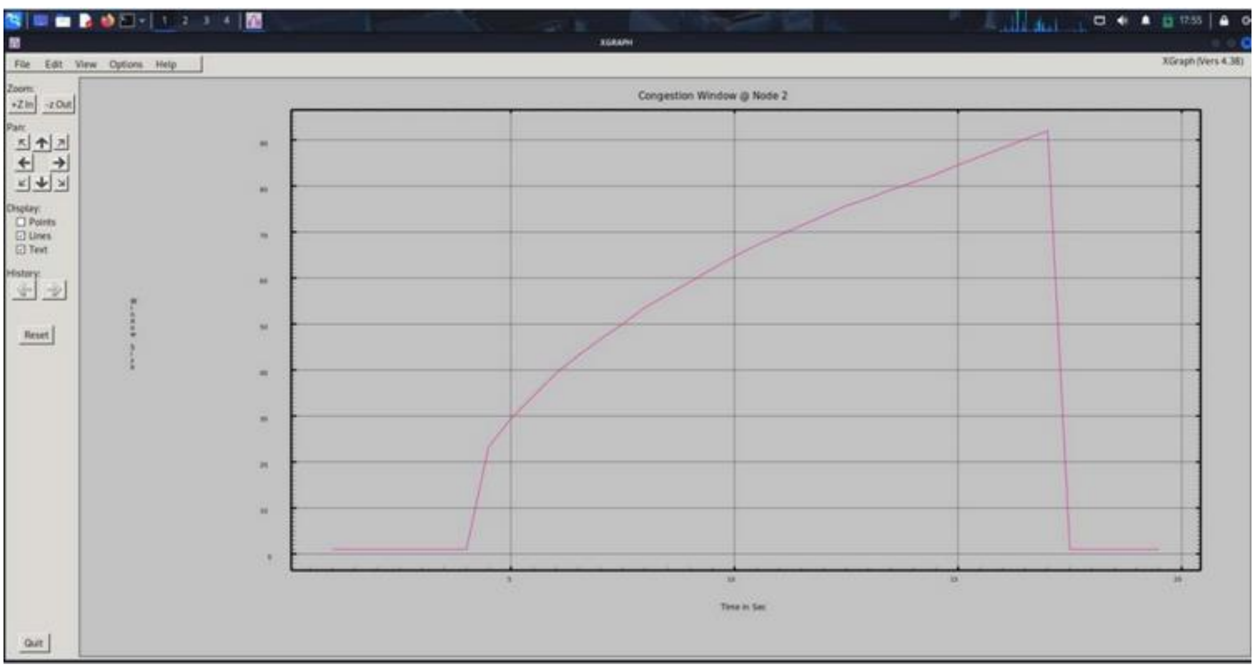
At around 10 seconds, the congestion window size drops sharply. This sudden decrease typically indicates packet loss or network congestion.

Congestion Avoidance & Recovery (10 - 20 seconds):

After the drop, the CWND fluctuates at a lower value, slowly increasing again. This phase represents TCP's *congestion avoidance* mechanism, where the sender cautiously increases the window size in a linear fashion to prevent further congestion.

Another Congestion Event (~20 seconds):

At around 20 seconds, another sharp drop in the congestion window occurs, suggesting a second instance of congestion or packet loss.



Explanation:

X-axis (Time in Sec): Represents time in seconds.

Y-axis (Window Size): Represents the size of the TCP congestion window, dictating how many packets can be sent before waiting for acknowledgment.

Slow Start (0 - ~3 seconds):

During this phase, the congestion window remains small initially, then starts to increase exponentially. This indicates TCP’s rapid growth in the window size as part of the slow start phase.

Congestion Avoidance (~3 - 15 seconds):

Here, the window size grows more gradually, reflecting a linear increase as TCP transitions into *congestion avoidance* mode to control network congestion.

Sudden Drop (~15 seconds):

At around 15 seconds, the congestion window experiences a sharp drop from its peak, suggesting a packet loss event caused by congestion or a timeout.

Stabilization (~15 - 20 seconds):

After the drop, the CWND remains at a low value for a while. This indicates that TCP is waiting for retransmissions, and the window size starts to increase again slowly.

TCL File


File Actions Edit View Help
GNU nano 8.3
Lab3_CW_Updated.tcl
Simulation parameters setup
set val(stop) 20.0 ;# time of simulation end
Initialization
#Create a ns simulator
set ns [new Simulator]
#Open the NS trace file
set tracefile [open out1.tr w]
$ns trace-all $tracefile
#Open the NAM trace file
set namfile [open out.nam w]
$ns namtrace-all $namfile
set file6 [open cw6.out w]
set file2 [open cw2.out w]
Nodes Definition
#Create 7 nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
set n6 [$ns node]
Links Definition
#Createlinks between nodes
$ns duplex-link $n0 $n1 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n1 100


GNU nano 8.3
Lab3_CW_Updated.tcl
$ns queue-limit $n0 $n1 100
$ns duplex-link $n0 $n2 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n2 50
$ns duplex-link $n0 $n3 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n3 50
$ns duplex-link $n0 $n4 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n4 50
$ns duplex-link $n1 $n5 100.0Mb 10ms DropTail
$ns queue-limit $n1 $n5 50
$ns duplex-link $n1 $n6 100.0Mb 10ms DropTail
$ns queue-limit $n1 $n6 50
Agents Definition
set tcp0 [new Agent/TCP]
set sink0 [new Agent/TCPSink]
$tcp0 set packetSize_ 3000
set tcp1 [new Agent/TCP]
set sink1 [new Agent/TCPSink]
$tcp1 set packetSize_ 1500
set udp0 [new Agent/UDP]
set sink1 [new Agent/LossMonitor]
$udp0 set packetSize_ 1500
Applications Definition
set ftp0 [new Application/FTP]
$ftp0 attach-agent $tcp0
set ftp1 [new Application/FTP]
$ftp1 attach-agent $tcp1
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next
Back
Forward
Prev Word
Next Word
Home
End

File Actions Edit View Help

```
 GNU nano 8.3                                        Lab3_CW_Updated.tcl *
proc record {} {
        global tcp0 tcp1 file6 file2

        #Get an instance of the simulator
        set ns [Simulator instance]

        #Set the time after which the procedure should be called again
        set time 0.5

        #How many bytes have been received by the traffic sinks?
        set cw6 [$tcp0 set cwnd_]
        set cw2 [$tcp1 set cwnd_]

        #Get the current time
        set now [$ns now]

        puts $file6  $now $cw6
        puts $file2  $now $cw2

        #Re-schedule the procedure
        $ns at [expr $now+$time] [illegible]
}

$ns at 1.0 [illegible]
$ns at 2.0  $ftp0 [illegible]
$ns at 4.0  $ftp1 [illegible]
$ns at 10.0  $cbr0 [illegible]
$ns at 15.0  $cbr0 [illegible]
$ns at 17.0  $ftp1 [illegible]
$ns at 19.0  $ftp0 [illegible]


#===================================
#        Termination
#===================================
#Define a 'finish' procedure
proc finish {} {
    global ns namfile tracefile file6 file2
    $ns flush-trace
    close $tracefile
    close $namfile
    close $file6
    close $file2
    exec nam out.nam &
    exec /home/thilak/Documents/CN/XGraph4.38_linux64/bin/xgraph  cw6.out &
    exec /home/thilak/Documents/CN/XGraph4.38_linux64/bin/xgraph  cw2.out &
    exit 0
}
$ns at $val(stop) [illegible]
$ns run
```